

GLADSTONE RADAR, Australia

WMO: 943800

Lat: 23.8553S

Lon: 151.2628E

Elev: 75

StdP: 100.42

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	12.7	-2.6	3.1	16.7	-0.1	3.8	16.5	10.0	20.7	9.0	20.2	3.8	220	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	32.8	24.9	31.9	24.5	31.1	24.0	26.6	30.7	25.9	29.5	25.3	28.9	5.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.0	28.7	25.0	20.2	28.1	24.4	19.5	27.5	83.9	30.9	80.9	29.5	78.3	28.8	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.6	8.7		9.0	35.4	1.1	1.5	8.2	36.6	7.5	37.5	6.9	38.3	6.1	39.4
				4.6	27.8	0.7	1.0	4.1	28.6	3.6	29.2	3.2	29.7	2.7	30.5
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	26.9	26.6	25.7	23.9	21.3	18.9	18.5	19.3	21.7	23.3	25.1	26.3
	DBStd	3.46	1.63	1.68	1.51	1.55	1.77	2.08	1.82	1.86	1.67	1.75	1.81	1.87
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	53	0	0	0	0	2	18	21	12	1	0	0	0
	CDD10.0	4782	523	465	488	416	349	268	263	288	350	414	453	505
	CDD18.3	1793	265	232	229	166	93	36	25	42	101	156	203	247
	CDH23.3	11686	2360	1998	1659	805	247	47	34	87	360	745	1323	2019
	CDH26.7	3202	766	626	439	136	13	1	1	3	39	141	378	659
Wind	WSAvg	4.7	5.5	5.5	5.3	5.0	4.3	4.1	3.9	4.0	4.2	4.7	4.9	5.1
Precipitation	PrecAvg	929	147	155	101	47	57	37	35	32	24	56	72	133
	PrecMax	1732	640	710	371	250	316	220	170	132	130	277	218	509
	PrecMin	432	6	7	12	0	0	0	0	0	0	3	1	3
	PrecStd	290	129	162	95	51	65	45	43	31	27	55	50	114
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	33.7	32.5	30.6	28.0	26.0	25.7	27.0	29.5	30.9	32.4	33.9
		MCWB	25.7	25.5	24.7	23.3	20.2	19.5	17.7	18.7	20.2	22.1	23.3	24.8
	2%	DB	32.7	32.5	31.2	29.1	26.7	24.6	24.3	25.3	27.7	29.3	31.0	32.4
		MCWB	25.2	25.2	24.1	22.3	19.8	18.6	17.4	17.9	19.7	22.0	23.1	24.4
	5%	DB	31.5	31.4	30.2	28.2	25.7	23.6	23.2	24.2	26.6	28.2	30.0	31.3
		MCWB	24.5	24.8	23.6	21.6	19.1	17.8	16.7	17.5	19.3	21.3	22.7	23.9
	10%	DB	30.5	30.3	29.2	27.3	24.8	22.6	22.1	23.1	25.5	27.2	29.0	30.2
		MCWB	24.0	24.1	23.0	21.0	18.7	17.3	16.1	16.8	18.8	20.7	22.1	23.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.4	26.5	25.4	23.1	21.5	20.8	21.2	23.1	24.5	26.0	27.1
		MCDB	31.9	31.7	30.1	28.2	25.6	23.8	22.9	24.5	26.7	28.8	30.4	31.8
	2%	WB	26.5	26.5	25.6	24.0	22.1	20.3	19.6	20.1	21.9	23.5	24.8	26.0
		MCDB	30.3	30.5	28.8	27.2	24.2	22.7	21.9	23.0	25.3	27.3	28.9	29.9
	5%	WB	25.8	25.7	24.9	22.9	21.0	19.4	18.7	19.2	21.1	22.7	24.0	25.2
		MCDB	29.0	29.3	28.2	26.3	23.9	21.9	20.9	21.9	24.3	26.1	27.8	28.9
	10%	WB	25.1	25.1	24.2	22.1	20.1	18.6	17.8	18.4	20.3	22.0	23.2	24.6
		MCDB	28.5	28.5	27.6	25.7	23.2	21.2	20.4	21.3	23.5	25.3	26.9	28.2
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.2	7.2	7.4	7.6	7.4	8.0	8.6	8.7	8.2	8.0	7.7
		MCDBR	8.7	8.6	8.4	8.3	8.3	8.0	8.7	9.4	9.6	9.2	8.9	9.0
		MCWBR	3.3	3.4	3.5	3.5	4.0	4.0	4.5	4.4	4.2	3.8	3.5	3.6
	5% WB	MCDBR	7.6	7.7	7.2	7.4	7.0	6.9	7.4	8.0	8.4	8.3	8.2	8.1
		MCWBR	3.5	3.5	3.3	3.3	3.5	3.6	3.9	4.3	4.2	4.0	3.8	3.6
Clear-Sky Solar Irradiance	taub	0.389	0.386	0.372	0.344	0.322	0.315	0.305	0.316	0.353	0.374	0.391	0.391	
	taud	2.530	2.547	2.570	2.638	2.646	2.662	2.667	2.622	2.508	2.485	2.463	2.502	
	Ebn at Noon	955	944	930	916	898	885	908	931	934	945	947	955	
	Edh at Noon	112	108	101	87	79	75	76	87	106	114	119	116	
All-Sky Solar Radiation	RadAvg	6.62	6.14	5.48	4.81	4.08	3.53	4.00	4.86	5.90	6.52	6.88	6.88	
	RadStd	0.46	0.64	0.46	0.40	0.23	0.28	0.30	0.33	0.44	0.46	0.64	0.66	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page